

Expt. No. : 2
Date :

BUILD A PYTHON PROGRAM TO GET A RANGE OF NUMBERS FROM USER AND TO SEPARATE EVEN NUMBERS AND ODD NUMBERS RESPECTIVELY

AIM

To write a Python program to separate even and odd numbers from the given range of numbers.

SOFTWARE REQUIRED

Python 3.14

FORMULA USED

Even Numbers:

n is even if $n \% 2 == 0$

$n \% 2$ calculates the remainder when n is divided by 2. If the remainder is 0, n is even.

Odd Numbers:

n is odd if $n \% 2 \neq 0$

If the remainder is not 0 when n is divided by 2, n is odd.

THEORY

Even Numbers: An integer is classified as even if it is divisible by 2 without any remainder. Mathematically, a number n is even if there exists an integer k such that:

$$n = 2K$$

Examples of even numbers include 2, 4, 6, 8, 10, etc

Odd Numbers: An integer is classified as odd if it is not divisible by 2 without any remainder. In other words, when divided by 2, it leaves a remainder of 1.

Mathematically, a number n is odd if there exists an integer k such that:

$n = 2k + 1$ Examples of odd numbers include 1, 3, 5, 7, 9, etc.

PROBLEM

The electricity meter numbers are categorized into two groups according to their numerical property. Write a Python program to separate the meter numbers from a given range.

APPLICATION PROBLEM

The sensor node numbers in an electrical monitoring system are numbered consecutively. To organize data collection efficiently, the sensor nodes are classified into two groups based on their numbering pattern. Write a Python program to separate the sensor numbers from a given range.

ALGORITHM

STEP 1: Start

STEP 2: Input: Lower bound (LB) and Upper bound (UB) for the range of numbers

STEP 3: Initialize two empty lists: EvenNumbers and OddNumbers

STEP 4: Loop through each number from LB to UB:

a. If the number modulo 2 equals 0:

i. Append the number to EvenNumbers list

b. Else:

i. Append the number to OddNumbers list

STEP 5: Construct a statement using the EvenNumbers and OddNumbers lists:

STEP 6: If EvenNumbers list is not empty:

i. Construct a statement listing even numbers

STEP 7: If OddNumbers list is not empty:

i. Construct a statement listing odd numbers

STEP 8: Output the constructed statement

STEP 9: End

PROGRAM CODING

```

n = int(input("Enter the number of meter numbers: "))
even = []
odd = []
for i in range(n):
    num = int(input("Enter the meter number: "))
    if num % 2 == 0:
        even.append(num)
    else:
        odd.append(num)
print("Even meter numbers:", even)
print("Odd meter numbers:", odd)

```

OUTCOMES OF PROGRAM CODING

```

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment2.py =====
Enter the number of meter numbers: 5
Enter the meter number: 56
Enter the meter number: 248
Enter the meter number: 546
Enter the meter number: 5484
Enter the meter number: 89
Even meter numbers: [56, 248, 546, 5484]
Odd meter numbers: [89]
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment2.py =====
Enter the number of meter numbers: 4
Enter the meter number: 544
Enter the meter number: 12334
Enter the meter number: 68
Enter the meter number: 412
Even meter numbers: [544, 12334, 68, 412]
Odd meter numbers: []
>>>

```

APPLICATION PROBLEM CODING

```

n = int(input("Enter the number of sensor numbers: "))
even = []
odd = []
for i in range(n):
    num = int(input("Enter the sensor number: "))
    if num % 2 == 0:
        even.append(num)
    else:
        odd.append(num)
print("Even sensor numbers:", even)
print("Odd sensor numbers:", odd)

```

OUTCOMES OF APPLICATION PROGRAM CODING

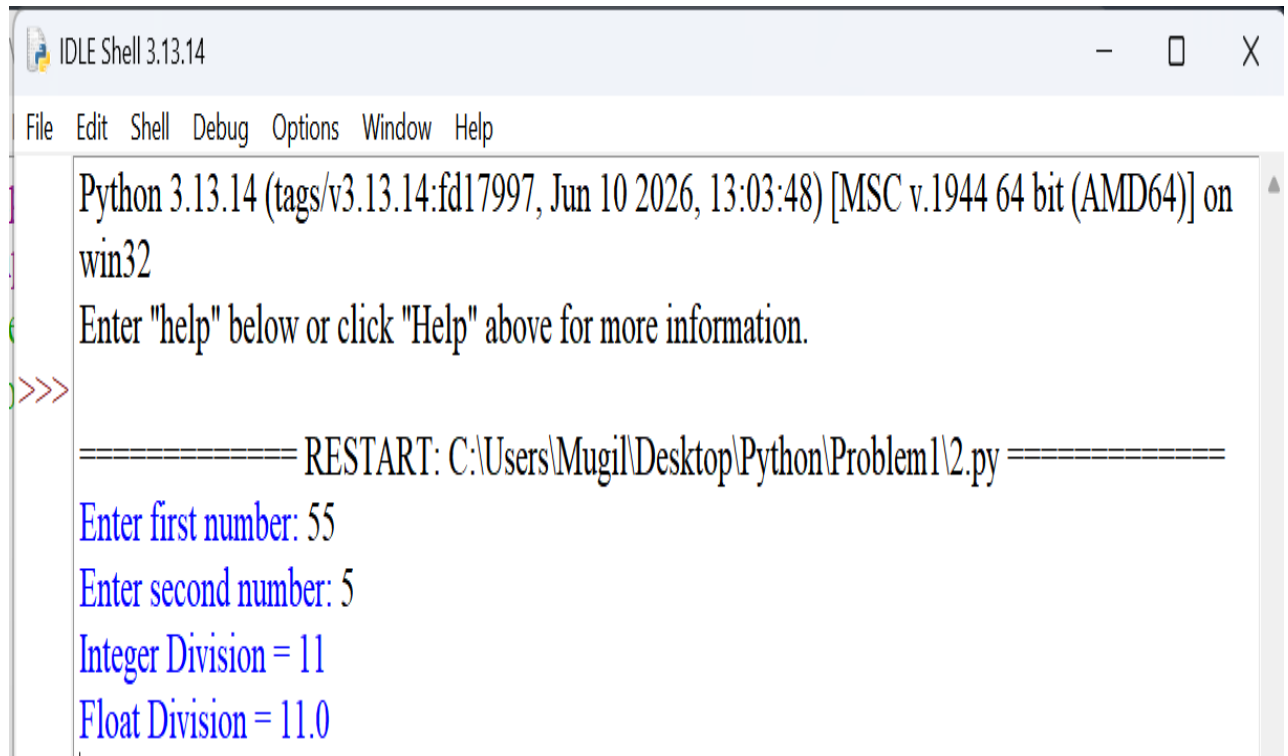
```

IDLE Shell 3.13.14
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Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment2.py =====
Enter the number of sensor numbers: 5
Enter the sensor number: 1355
Enter the sensor number: 4654
Enter the sensor number: 67
Enter the sensor number: 5546
Enter the sensor number: 165
Even sensor numbers: [4654, 5546]
Odd sensor numbers: [1355, 67, 165]
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment2.py =====
Enter the number of sensor numbers: 5
Enter the sensor number: 4569
Enter the sensor number: 1336
Enter the sensor number: 4668
Enter the sensor number: 45566
Enter the sensor number: 123
Even sensor numbers: [1336, 4668, 45566]
Odd sensor numbers: [4569, 123]
>>> |

```

PROGRAM CODING

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
print("Integer Division =", a // b)  
print("Float Division =", a / b)
```

OUTCOMES OF PROGRAM CODING

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem1\2.py =====  
Enter first number: 55  
Enter second number: 5  
Integer Division = 11  
Float Division = 11.0
```

VIVA QUESTION**1. How are arguments passed by value or by reference in Python?**

In Python, arguments are passed by object reference. Mutable objects (e.g., lists) can be changed within a function, while immutable objects (e.g., integers, strings) cannot be altered.

2. What is the difference between a Set and Dictionary?

Set: Unordered collection of unique elements. Example: {1, 2, 3}.

Dictionary: Unordered collection of key-value pairs. Example: {'a': 1, 'b': 2}.

3. What is List Comprehension? Give an Example.

List comprehension is a concise way to create lists using a single line of code.

Example:

```
squares = [x**2 for x in range(5)] # Output: [0, 1, 4, 9, 16]
```

4. What is a lambda function?

A lambda function is an anonymous, one-line function defined with the lambda keyword.

Example:

```
add = lambda x, y: x + y print(add(2, 3)) # Output: 5
```

5. What is a pass in Python?

The pass statement is a placeholder used in empty code blocks where code is syntactically required but no action is needed.

Example:

```
def my_function(): pass # No code yet
```

RESULT:

Thus, the Python program successfully separated the odd and even numbers, from the given range were executed correctly, and verified the separation. Also, its application program was also executed and verified successfully.

INFERENCE:

The Python program correctly classified odd and even numbers using conditional statements and loops. The output verified the program's accuracy, demonstrating Python's effectiveness in application program by solving basic computational problems successfully.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		